

Orders, Returns & Profitability

Revenue • Customer Lifecycle • Customer Value • Products • Returns

Hady Abohany

Revenue & Margins

Yousef Sayed

Customer Lifecycle

Ahmed Mohamed

Customer Value

Ibrahim Desouky

Products & Devices

Ahmed Shaban

Products & Devices

Rojeh Tamer

Power BI Dashboard

Team & Responsibilities

Work distribution for the project

Name	Responsibility
Hady Abohany	Revenue & Margins • Visualization
Yousef Sayed	Customer Lifecycle
Ahmed Mohamed	Customer Value
Ibrahim Desouky	Customer Value + first three product / device queries
Ahmed Shaban	Worked with Ibrahim on the first three product / device queries
Rojeh Tamer	Built the interactive dashboard in Power BI

DASHBOARD

Power BI

Interactive dashboard built by Rojeh Tamer

ANALYSIS

SQL

Queries for revenue, lifecycle, value, products & returns

TEAM SIZE

6

Members across four analysis areas

Project Overview

Turning marketplace data into operational insights

- E-commerce marketplace operating across multiple regions and devices.
- Customers place orders, purchase items, and sometimes return them.
- The analysis answers the COO's questions about revenue, customer behavior, products and returns.

Category	Value	Details
Users	3,000	p2_users table
Core tables	4	users • orders • order_items • returns
Analysis areas	4	revenue • lifecycle • value • products/returns

Data Structure

Project 2 tables used in the analysis

Table name	Key columns
p2_users	user_id • region • signup_date
p2_orders	order_id • user_id • order_date • device
p2_order_items	order_id • sku • category • qty • price
p2_returns	order_id • return_reason • return_date

CATEGORIES

ELEC • HOME • FASH • SPORT • BEAUTY • GROC

Analytical Workflow

SQL queries → results → visualizations → dashboard

01

Query

SQL aggregates and calculations

02

Measure

Revenue, margin, retention,
spend, returns

03

Visualize

Charts from the notebook and
Power BI

04

Interpret

Business takeaways for the COO

REVENUE AFTER DISCOUNT

Σ (GMV - Discount)

Used for monthly revenue analysis

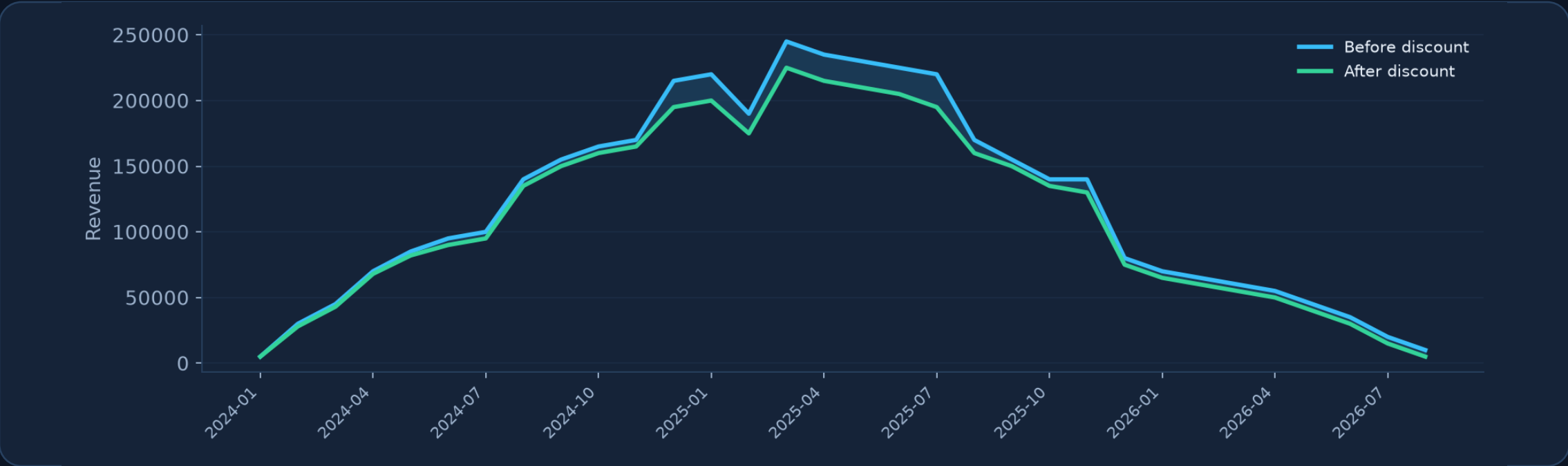
MARGIN

GMV - Discount - COGS - Shipping

Used for regional profitability

Revenue & Margins

Hady • Monthly revenue before vs. after discounts



PEAK MONTH

Mar 2025

Revenue before discount: 245,403.20

DISCOUNT EFFECT

Visible gap

After-discount revenue stays below gross GMV

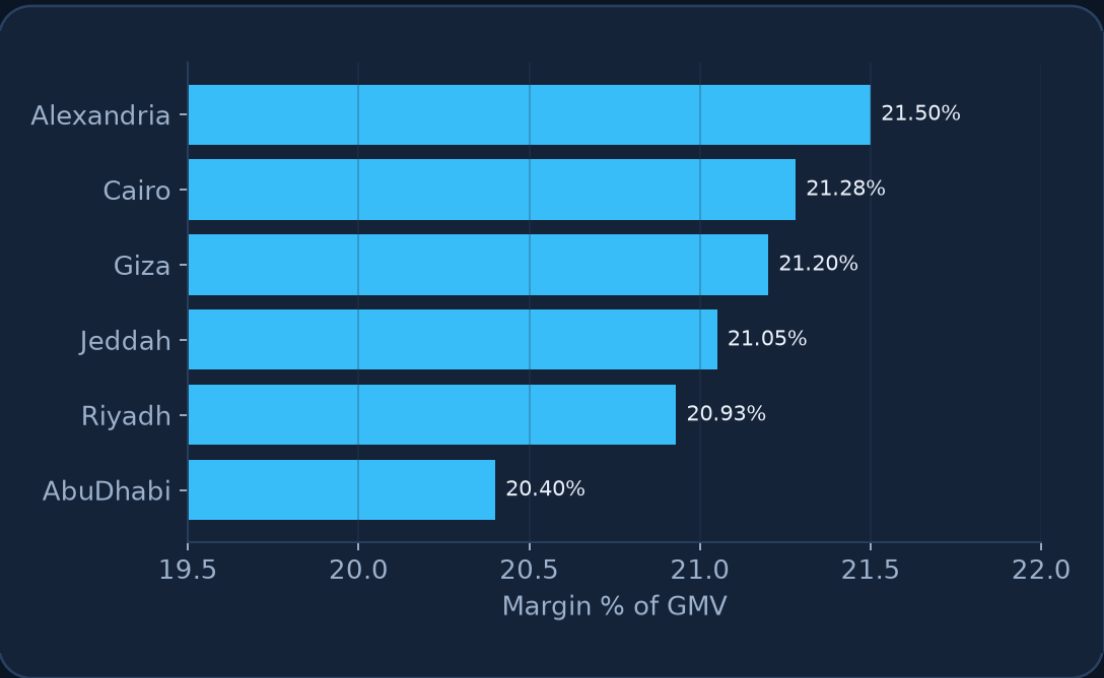
TREND

Growth → decline

Rises into early 2025, declines afterwards

Regional Margins

Profitability after discounts, COGS and shipping



Region	Margin % of GMV
Alexandria	21.50%
Cairo	21.28%
Giza	21.20%
Jeddah	21.05%
Riyadh	20.93%
AbuDhabi	20.40%

MARGIN RANGE

20.40% - 21.50%

Regional percentages are tightly grouped

HIGHEST

Alexandria • 21.50%

Highest regional margin in the output

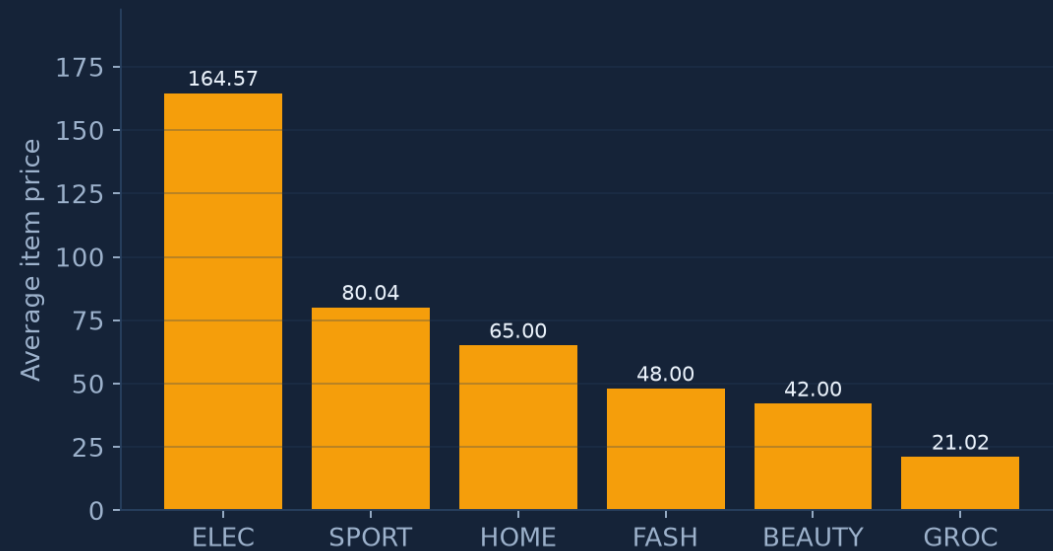
LOWEST

AbuDhabi • 20.40%

Lowest regional margin in the output

Average Item Price by Category

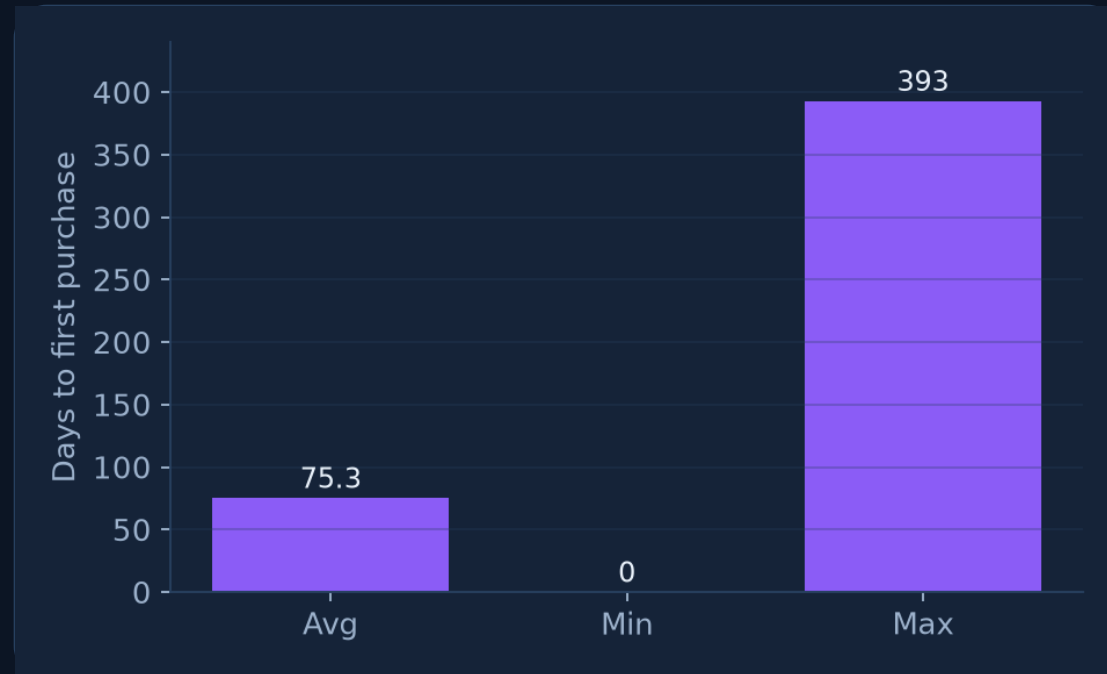
Hady • supporting product economics



Category	Average price
ELEC	164.57
SPORT	80.04
HOME	65.00
FASH	48.00
BEAUTY	42.00
GROC	21.02

Customer Lifecycle

Yousef • From signup to repeat purchase



AVG FIRST PURCHASE

75.3 days

From signup to first order

REPEAT RATE

96.64%

Customers with more than one order

AVG 1ST → 2ND

72.8 days

Gap between first and second order

Customer Lifecycle — Interpretation

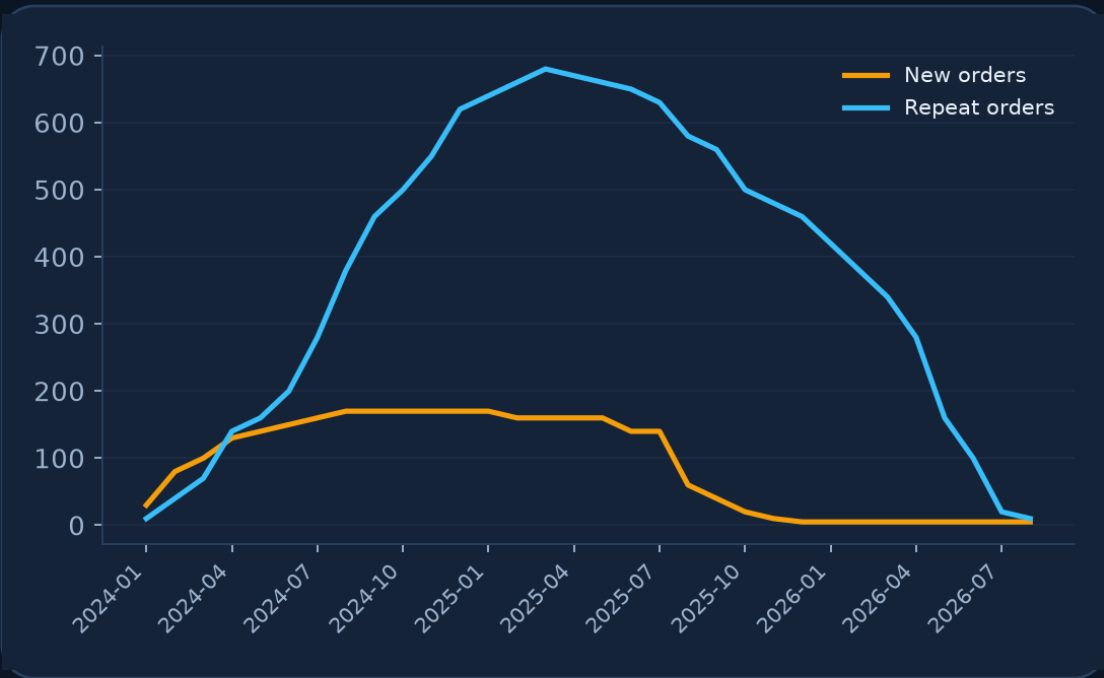
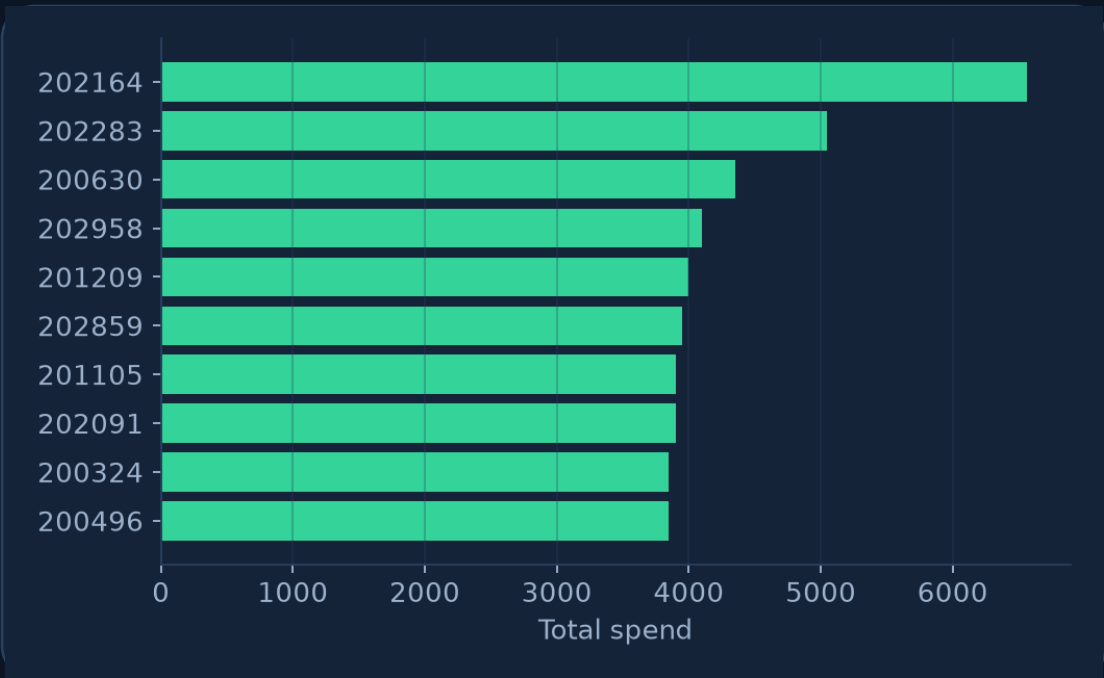
Key retention timing signals

- Average time from signup to first purchase: 75.3 days.
- 2,873 of 2,973 customers are classified as repeat customers.
- Average gap between first and second order: 72.8 days.
- The lifecycle analysis focuses on acquisition-to-purchase timing and repeat behavior.

Measure	Value	Description
First purchase	75.3 days	Average across customers with an order
Repeat customers	2,873	Customers with more than one order
Second order gap	72.8 days	Average time between first and second order

Customer Value

Ahmed Mohamed • Spend, order mix & recency



TOP CUSTOMER

202164 • 6,564.20

Highest total spend in the output

TOP CUSTOMER ORDERS

13

Orders placed by the top spender

REPEAT ORDERS

Overtake new orders

Around mid-2024 repeat passes new

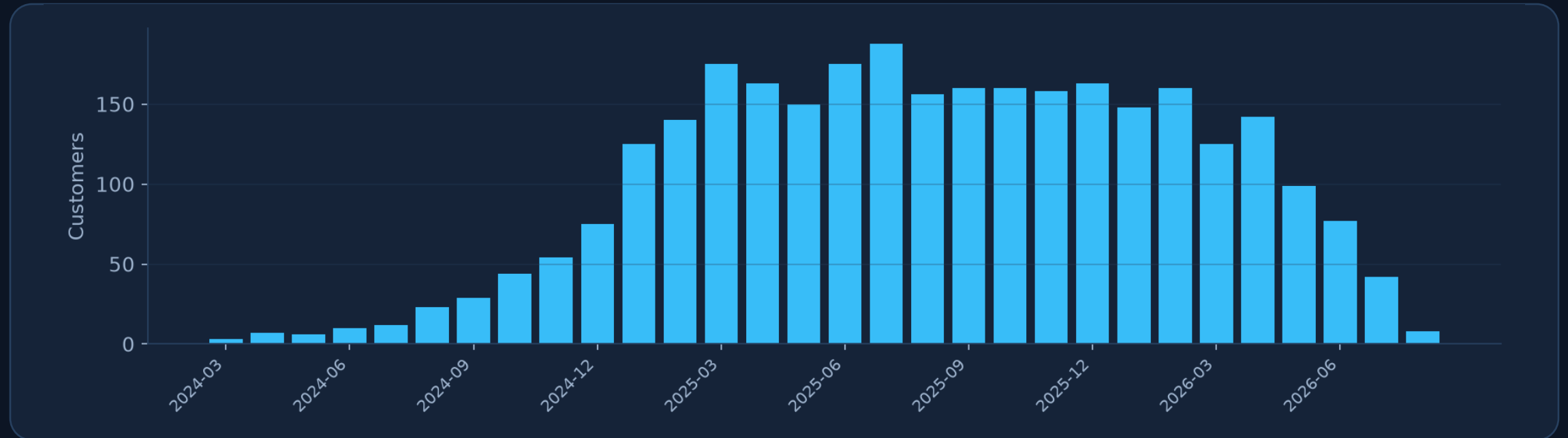
Top 10 Customers by Total Spend

Ahmed Mohamed • query output

#	User ID	Total spend
1	202164	6,564.20
2	202283	5,050.00
3	200630	4,350.00
4	202958	4,100.00
5	201209	4,000.00
6	202859	3,950.00
7	201105	3,900.00
8	202091	3,900.00
9	200324	3,850.00
10	200496	3,850.00

Customer Recency

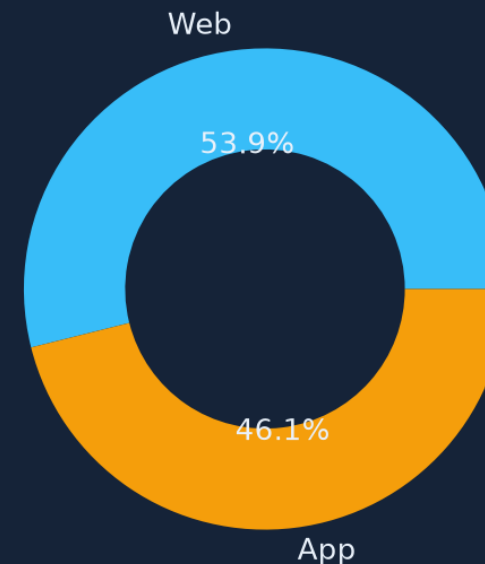
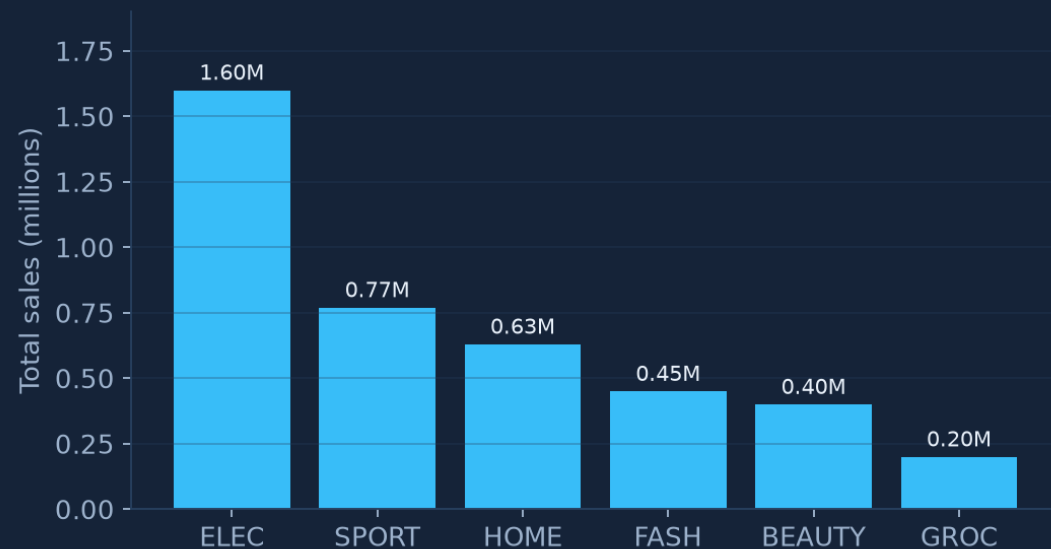
Most recent purchase month across customers



- The dataset ends in August 2026.
- Many customers' last purchases fall in the more recent months.
- A 60-90 day inactivity rate needs a follow-up calculation using the maximum order date.

Products & Devices

Ibrahim Desouky + Ahmed Shaban • first three queries



TOP SALES CATEGORY

ELEC • 1,594,473.44

Largest reported category sales

DEVICE SPLIT

Web 8,086 • App 6,914

About 54% / 46% of orders

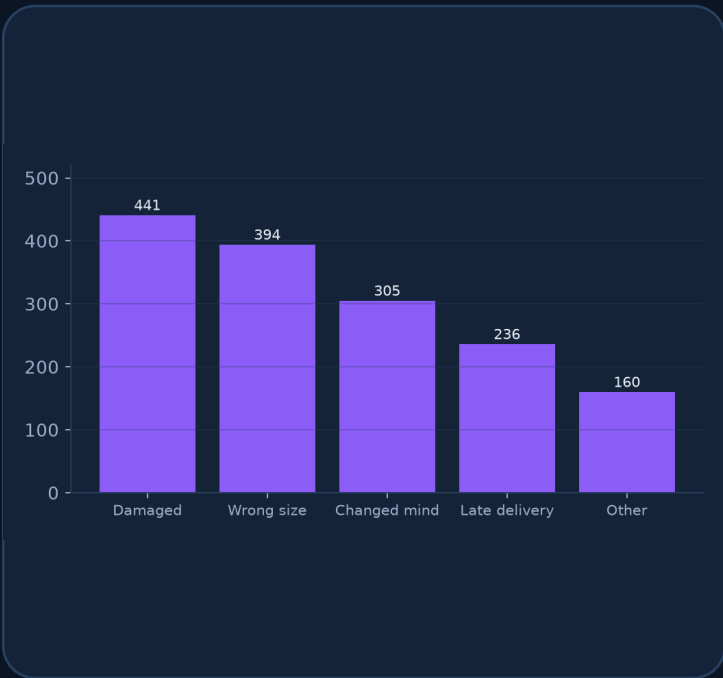
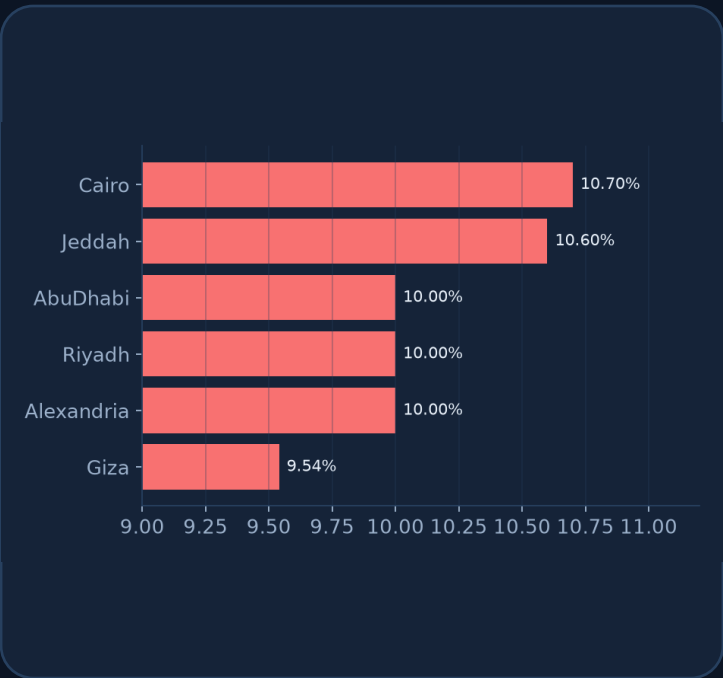
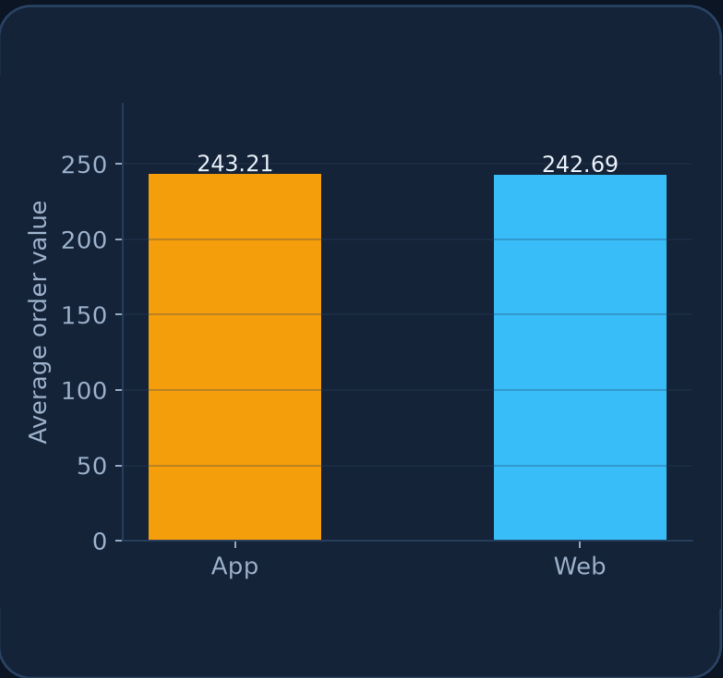
AVG ITEMS / ORDER

3.0

Order-level quantity indicator

Device Spend & Returns

Additional product and customer-experience analysis



AVG ORDER VALUE

App 243.21 • Web 242.69

Very close across devices

RETURN RATE RANGE

9.54% - 10.70%

Regional rates are relatively close

TOP RETURN REASON

Damaged • 441

Wrong size follows with 394

Executive Dashboard

A compact view of the main results

USERS

3,000

ORDERS

15,000

REPEAT CUSTOMERS

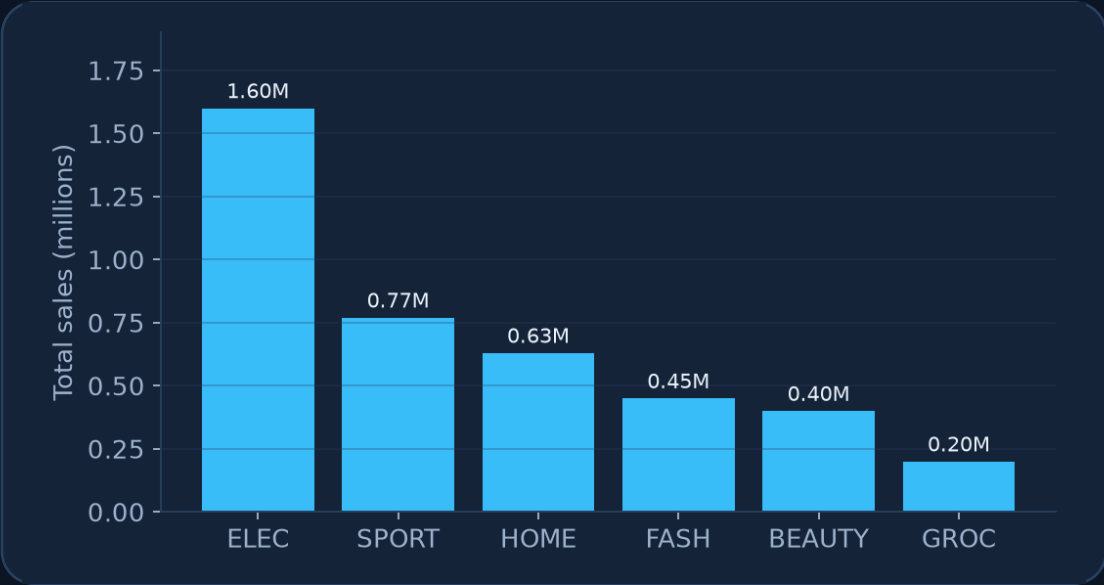
96.64%

AVG ITEMS / ORDER

3.0

HIGHEST RETURN RATE

10.70%



Key Findings

What the analysis shows

- Revenue increased substantially through early 2025 before declining later in the period.
- Regional margin percentages are tightly grouped around 20.4% – 21.5% of GMV.
- Repeat customers represent 96.64% of customers with orders.
- ELEC is the largest sales category and has the highest average item price.
- Web generates more orders than App, while average order value is nearly identical.
- Return rates are close across regions; damaged items and wrong size lead the reasons.

REVENUE

Trend + peak + decline

CUSTOMERS

Strong repeat behavior

RETURNS

Damage + sizing

Query Results — KPI Summary

A compact table of the main outputs

Metric	Result	What it tells us
Users	3,000	Customer base in p2_users
Orders	15,000	Orders analyzed
Customers with orders	2,973	Customers included in lifecycle analysis
Repeat customers	2,873	Customers with more than one order
Repeat-customer rate	96.64%	Repeat customers / customers with orders
Average items / order	3.0	SQL query result
Peak revenue month	Mar 2025	Highest monthly gross revenue

Query Results — Regional Margins

Margin = GMV - Discount - COGS - Shipping Cost

Region	Margin %
Alexandria	21.50%
Cairo	21.28%
Giza	21.20%
Jeddah	21.05%
Riyadh	20.93%
AbuDhabi	20.40%

MARGIN RANGE

20.40% - 21.50%

Regional percentages are relatively close.

HIGHEST REPORTED

Alexandria • 21.50%

Highest regional margin in the query output.

LOWEST REPORTED

AbuDhabi • 20.40%

Lowest regional margin in the query output.

Query Results — Customer Lifecycle

Signup → first purchase → repeat purchase timing

Lifecycle measure	Result
Average days to first purchase	75.3 days
Minimum days to first purchase	0 days
Maximum days to first purchase	393 days
Customers with orders	2,973
Repeat customers	2,873
Repeat-customer rate	96.64%
Average gap first → second order	72.8 days

READING THE RESULT

Most customers in the analysis are repeat customers; the timing metrics quantify the purchase cycle.

Query Results — Customer Value

Spend and order-frequency outputs

Measure	Result
Top customer by spend	202164 • 6,564.20
Orders by the top customer	13
Second highest spend	202283 • 5,050.00
Repeat vs new orders	Repeat overtakes new around mid-2024

TOP SPENDER

202164

6,564.20 total spend

ORDER COUNT

13

Same top-spend customer

ORDER MIX

Repeat > New

Around mid-2024

Query Results — Products & Categories

Category sales and average item-price outputs

Category / KPI	Result	Note
ELEC	1,594,473.44	Highest reported category sales
SPORT	≈ 770,000	Second category by sales
HOME	≈ 630,000	Third category by sales
Highest average item price	ELEC • 164.57	Highest reported average price
Lowest average item price	GROC • 21.02	Lowest reported average price
Average items per order	3.0	Order-level quantity indicator

KEY OUTPUT

ELEC leads reported sales and also has the highest reported average item price.

Query Results — Device Performance

Order volume and average order value by device

Device	Orders	Avg order value
Web	8,086	242.69
App	6,914	243.21
Total	15,000	≈ 242.95

WEB

8,086 orders

More orders in the query output

APP

6,914 orders

Lower order count than Web

AVG ORDER VALUE

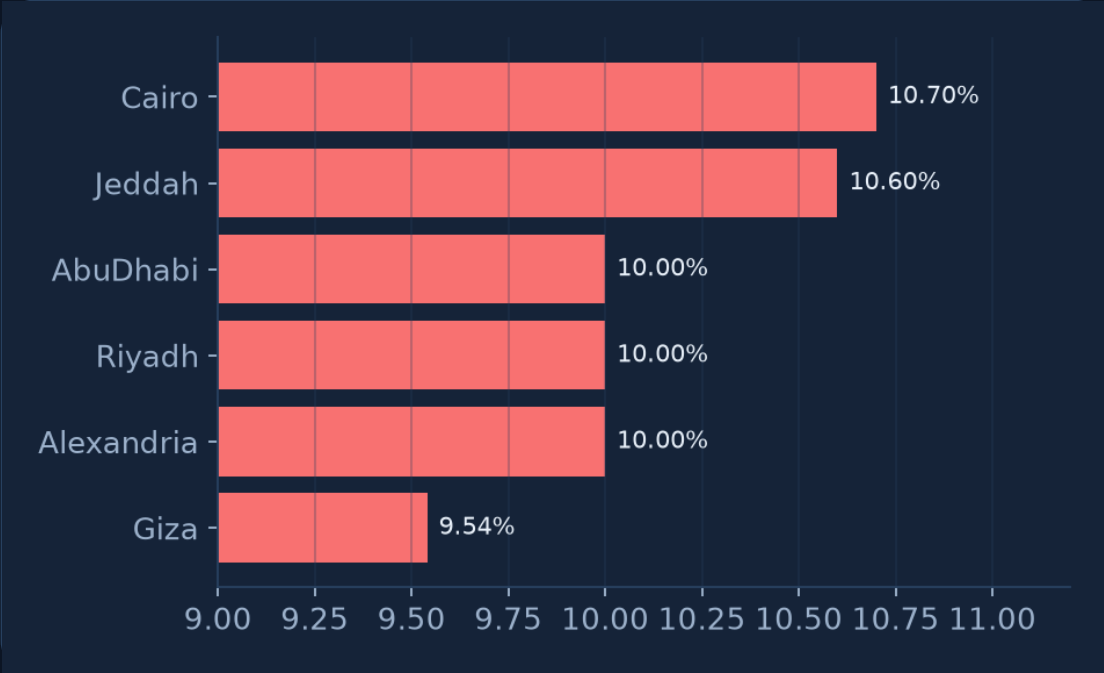
Nearly identical

Very close across devices

Query Results — Return Rate by Region

Regional return-rate outputs

Region	Return rate %
Cairo	10.70%
Jeddah	10.60%
AbuDhabi	10.00%
Riyadh	10.00%
Alexandria	10.00%
Giza	9.54%



HIGHEST

Cairo • 10.70%

Highest reported regional return rate

LOWEST

Giza • 9.54%

Lowest reported regional return rate

SPREAD

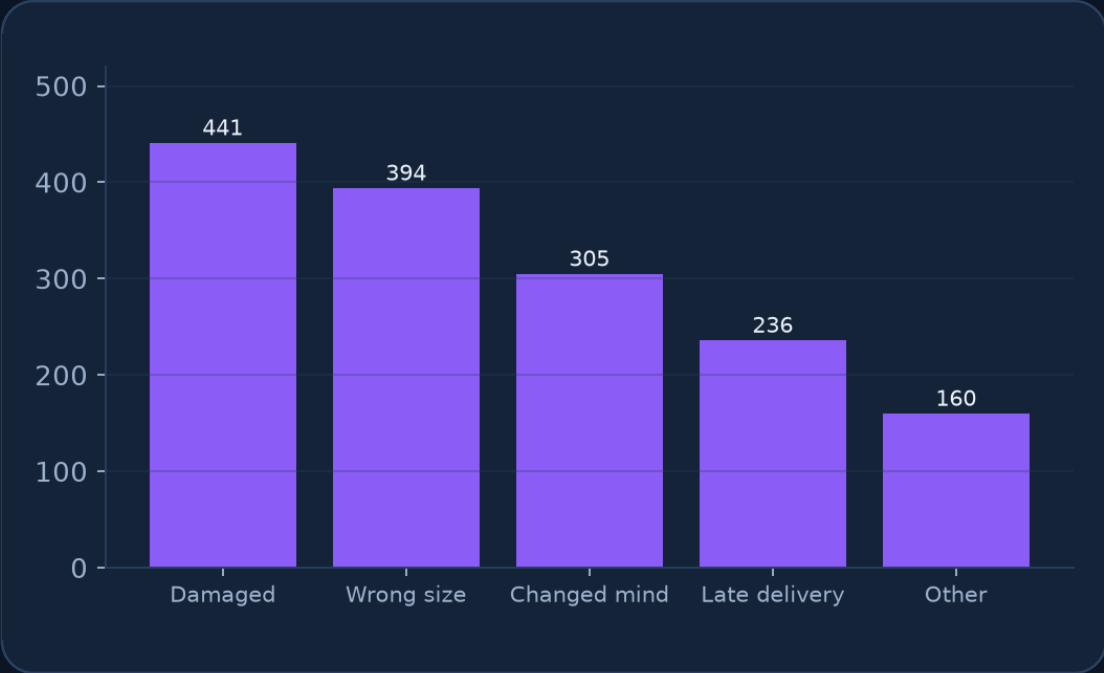
1.16 pts

The query shows no large regional spread

Query Results — Return Reasons

Most frequent return reasons in the reported results

Return reason	Count
Damaged	441
Wrong size	394
Changed mind	305
Late delivery	236
Other	160



TOP REASON

Damaged • 441

Highest frequency in the reported result

SECOND

Wrong size • 394

Second-highest frequency

TOTAL RETURNS LISTED

1,536

Sum of the five reported reasons

Interactive Power BI Dashboard

Built by Rojeh Tamer • dashboard screenshot as supplied



TOOL

Microsoft Power BI

Built by Rojeh Tamer

PAGES

Overview • Deep dive

Region, device, category and date filters

HEADLINE KPIS

666.27K revenue

163.00K margin • 606 orders • 1.10K AOV